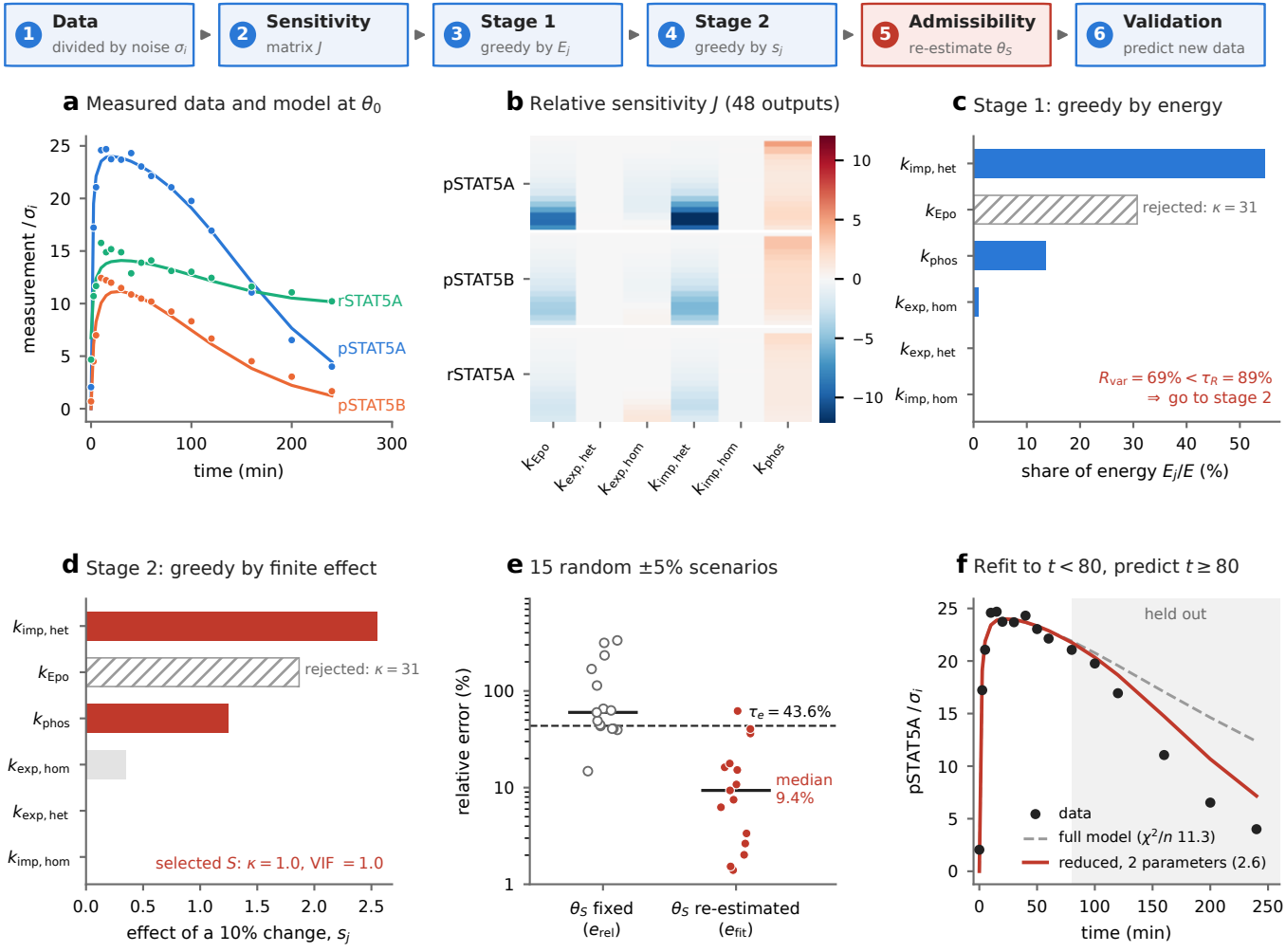
1 Data
divided by noise σ_i
2 Sensitivity
matrix J
3 Stage 1
greedy by E_j
4 Stage 2
greedy by s_j
5 Admissibility
re-estimate θ_S
6 Validation
predict new data
a Measured data and model at θ_0
measurement / σ_i
time (min)
rSTAT5A
pSTAT5A
pSTAT5B
b Relative sensitivity J (48 outputs)
pSTAT5A
pSTAT5B
rSTAT5A
k_{Epo}
$k_{exp,het}$
$k_{exp,hom}$
$k_{imp,het}$
$k_{imp,hom}$
k_{phos}
c Stage 1: greedy by energy
rejected: $\kappa = 31$
$R_{var} = 69\% < \tau_R = 89\%$
⇒ go to stage 2
share of energy E_j/E (%)
d Stage 2: greedy by finite effect
rejected: $\kappa = 31$
selected S: $\kappa = 1.0$, VIF = 1.0
effect of a 10% change, s_j
e 15 random ±5% scenarios
relative error (%)
$\tau_e = 43.6\%$
median 9.4%
θ_S fixed (e_{rel})
θ_S re-estimated (e_{fit})
f Refit to $t < 80$, predict $t \geq 80$
held out
pSTAT5A / σ_i
data
full model (χ^2/n 11.3)
reduced, 2 parameters (2.6)
time (min)